`llm-smart-router`

Multi-model AI agent orchestrator

June Gu · CMU 06-642 · Spring 2026

`github.com/junegu-glitch/llm-smart-router`

The Problem

Most AI workflows treat every task as input to one chat window.

But models have real specializations:

Task	Best fit	Why
Code generation	Claude Sonnet	Benchmark leader
Long documents	Gemini 2.5 Pro	1M token context
Math / reasoning	DeepSeek R1	54× cheaper than Claude
General / cheap	DeepSeek V3	$0.028 / 1M tokens

Switching manually is friction. Choosing wrong is waste.

The Solution

```
# Automatically routes to the right model
smart-router "Implement a binary search tree in Python"
→ Route: [coding] → Claude Sonnet 4

# Multi-model team for larger tasks
smart-router team run --preset code-review --git-diff "Review my changes"
→ SecurityReviewer + QualityReviewer + DocReviewer → Synthesis

# Zero cost with subscription CLIs
smart-router team run --use-cli "Compare React vs Vue"
→ Cost: $0.00
```

One command surface. Best model for each task. $0 with existing subscriptions.

Architecture

```
┌──────────────────────────────────────────────┐
│                User Input                    │
├──────────────────────────────────────────────┤
│   Router  (7-category classifier)            │
│   "write code" → coding → Claude Sonnet      │
├──────────────────────────────────────────────┤
│         Team Orchestrator                    │
│                                              │
│   ┌─────────┐   ┌─────────┐   ┌─────────┐    │
│   │ Claude  │   │ GPT-4o  │   │ Gemini  │  ← parallel
│   │[coding] │   │[writing │   │[research]│    │
│   └─────────┘   └─────────┘   └─────────┘    │
├──────────────────────────────────────────────┤
│   Leader synthesizes → Final Report          │
└──────────────────────────────────────────────┘
```

CLI layer (`src/cli/`) + Optional web UI (`src/app/`)

Key Innovation: $0 Team Mode

Most teams cost money per token. This one doesn't have to.

`--use-cli` **flag** — routes each teammate through locally installed subscription CLIs:

```
Claude teammate  → claude  CLI  (Anthropic subscription)
Gemini teammate  → gemini  CLI  (Google subscription)
Codex teammate   → codex   CLI  (OpenAI subscription)
```

If you already pay for these subscriptions, **team runs cost $0 in API tokens.**

```
smart-router team run --use-cli \
  "Compare Python vs Rust for a web scraper"

Total cost: $0.00   (3 models, 91 seconds)
```

Demo — CLI in Action

Demo — Team Mode

```
Smart Team — Multi-model agent team
✓ Leader planned: 3 teammates, 3 tasks

┌──────────────────────────────────────────────────────┐
│ Live Dashboard                  3/3 complete  97s    │
├──────────────────┬──────────────────┬──────────────────┤
│ Teammate         │ Model            │ Status  Cost     │
├──────────────────┼──────────────────┼──────────────────┤
│ TechnicalAnalyst │ Claude Sonnet    │ ✓ done  $0.031   │
│ PythonCoder      │ GPT-4.1          │ ✓ done  $0.001   │
│ RustCoder        │ Claude Sonnet    │ ✓ done  $0.062   │
└──────────────────┴──────────────────┴──────────────────┘

✓ Leader synthesized all results
Total cost: $0.094   Session saved: ~/.smart-router/sessions/
```

Web UI — Same Tool, Browser Interface

Live at: `https://scientific-software-engineering-wit.vercel.app`

Feature	Stack
Smart-routed chat	Same routing logic as CLI
Team Mode (`/team`)	SSE streaming, live dashboard, preset selector
GitHub OAuth login	Supabase Auth
API key management	AES-256-GCM encrypted, cloud-synced
Dark / light mode	Tailwind + system preference

CLI and web share the same `src/lib/` core — routing, model catalog, provider calls.

Team mode works both in `smart-router serve` ($0 with CLIs) **and** with BYOK API keys.

Evidence: Tests & CI

154 automated tests across 12 test files:

```
npm test  →  154 passed (154)  in 2.14s
```

Covered	Tests
Router (classification + model selection)	22
Team orchestration (plan → parallel → synthesize)	28
CLI hybrid mode (detection + subprocess)	21
Session save / load / list	18
Provider fallback chains	19

GitHub Actions CI — Node 20 + 22 matrix, runs on every push.

CI passing

Project Evolution

```
Project 1 (plateprep)
  └── Python CLI, narrow scope, one lab task
      No external APIs

Project 2 (llm-smart-router — CLI)
  └── TypeScript CLI, smart routing, team mode
      154 tests, provider integrations

Final Project (current)
  └── + GitHub Actions CI (Node 20 + 22)
      + --use-cli subscription hybrid ($0 cost)
      + Next.js web UI + Supabase auth + Vercel
      + Karpathy-style wiki (persistent context across sessions)
      + Public GitHub repo
```

Each step: same tool, deeper understanding of what "polished" means.

What I Learned About Agentic Engineering

Beginning of semester: AI writes draft → I fix it

By Project 2: Specify failure cases before implementation, not after

For final project:

- Plan mode before any significant change
- Wiki system for context across sessions (`/ingest`, `/query`, `/lint`)
- Stating invariants explicitly beats correcting output

The meta-moment: Used Claude Code to build a tool that runs Claude, Gemini, and Codex as parallel subprocesses.

> The orchestrator was built by an agent.
> The agents it orchestrates include the agent that built it.

Limitations & Future Work

- **Provider setup friction** — BYOK requires manual `config set` per provider; a guided `smart-router auth` wizard would lower the barrier
- **Routing edge cases** — mixed-type prompts (math + code) aren't classified well; ensemble classifier could improve precision
- **Cross-model verification** — the README describes Claude → Gemini → Codex → Claude judge pipeline, but it's not yet implemented as a preset
- **Web UI** — functional (OAuth, cloud sync, chat) but not polished enough for independent users

Thank you

`llm-smart-router` — route work to the best model, run parallel agent teams, $0 with subscription CLIs

`github.com/junegu-glitch/llm-smart-router`

```
npm install -g llm-smart-router
smart-router team run --use-cli "your complex task here"
```

Questions?